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Stochastic dynamic programming for hydropower reservoir operations with multiple local optima

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ABSTRACT

Multiple optima seriously confine the solution accuracy of stochastic dynamic programming (SDP) for non-concave maximization models of reservoir operation. To address the problem, a two-stage optimization algorithm combining traversing and search is proposed to obtain an optimal decision at each state combination. Single or multiple promising regions where a local optimal solution may exist are identified through coarse traversing in the whole feasible region, and a local search algorithm is used to local optimization in each promising region. Energy maximization model is used to analyze the multiple optima problem and test the method. In the model, quadratic and concave penalty functions are used to address the total power constraint for smaller shortages and fewer shortage periods respectively. The total power constraint increases the probability of multiple local optima existence and intensifies the difficulty in finding optimal decisions. The proposed model and method are applied to the cascaded reservoir system on middle-lower Lancang River in China. Numerical results show that the total power constraint is well addressed, the computing time can be reduced by more than 50% using the two-stage algorithm in getting operating rules with similar performance comparing to only traversing, and the multi-point search algorithm is superior to traversing and single point search in solving the proposed model.

1. Introduction

Stochastic dynamic programming (Loucks et al., 1981) is one of the most popular methods for reservoir operations. SDP can explicitly consider stream flow uncertainty in its recursive equation by solving the dynamic programming (DP) recursive relations adapted to stochastic problems, particularly Markov decision processes, and derive operating policies for different reservoir system states. SDP solutions have more computational demands than deterministic DP (Bai et al., 2015) as computation grows with the number of states, decisions and inflow scenario of reservoirs. Some amended or simplified methods to alleviate dimension problem were studied, such as the aggregation-disaggregation approaches (Turgeon and Charbonneau, 1998; Archibald et al., 1997), sampling SDP (SSDP) (Kelman et al., 1990; Faber and Stedinger, 2001) and stochastic dual dynamic programming (SDDP) (Pereira and Pinto, 1985; Pereira, 1989; Marques and Tilmant, 2013; Macian – Sorribes et al., 2017). Yeh (1985) and Labadie (2004) reviewed state-of-the-art of mathematical models for reservoir operations. Although many improvements have been made to SDP, it is still hard for SDP to solve large scale reservoir operation problems. Other approaches for reservoir operations include operating rule development applying deterministic optimization (Lund and Ferreira, 1996), simulation based optimization approaches (Koutsoyiannis and Economou, 2003; Chang et al., 2010; Liu et al., 2014; Giuliani et al., 2016), optimal reservoir trajectory (Turgeon, 2007; Wang, 2010) and reinforcement-learning (Busoniu et al., 2010; Castelletti et al., 2010).

Profit maximization, discounted (Weiner and Ben-Zvi, 1982) or undiscounted (Stedinger et al., 1984; Tejada-Guibert et al., 1993, 1995), is a common objective for reservoir systems. Some constraints can be easily addressed in SDP except the soft constraints such as lower bounds on reservoir release or power generation. Reliability or chance constrained model is a way to address the soft constraints. Askew (1974) proposed a chance-constrained dynamic programing (CCDP), in which the iterative search of a penalty for failure and a precise evaluation of system response were used within the dynamic program. Rossman (1977) presented an SDP model for determining reservoir release rules subjected to reliability constraints on system performance

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within the framework of Lagrangian duality theory. Sniedovich (1979) developed a dynamic programing solution for a class of reliability-constrained reservoir control problems with constraints defined over the entire life of the reservoir. Another way to deal with soft constraints is the penalty based methods. Stedinger et al. (1984) used quadratic penalty functions for monthly losses due to failure to meet flood control, irrigation and power generation targets of a single reservoir in SDP model using best inflow forecasting as a hydrologic state variable. Tejada-Guibert et al. (1993, 1995) used linear penalty function for water shortage and system power shortage in SDP models deriving multireservoir operating policies. In hydropower reservoir operation models, the lower bound on reservoir release is a linear constraint, and affects the solution slightly. While the total power constraint on a cascaded system with two or more reservoirs may lead to non-convex feasible regions even multiple local optima, and make the model much more difficult to solve.

A common solution approach for SDP is to test all the discretized decision points and find the best among them at each state, which is called traversing in this paper. The method does not require gradient evaluation and is robust but slow. Piccardi and Soncini-Sessa (1991) and Zhang et al. (2015) used parallel computation to improve solving efficiency of multi-dimensional DP. For objective functions with continuous second derivatives, more efficient methods such as quasi-Newton can be applied (Tejada-Guibert et al., 1993). Foufoula-Georgiou (1991) addressed interpolating nodal values and derivatives of a one-dimensional convex function with a convex interpolant so that potential computational difficulties from approximation-induced non-convexity are avoided, and an efficient convergence to global instead of local optimal controls is guaranteed at each single-stage optimization. Johnson et al. (1993) demonstrated that the computational effort required to develop numerical solutions to continuous-state dynamic programs can be reduced significantly when cubic piecewise polynomial functions, rather than tensor product linear interpolants, are used to approximate the value function. For problems with concave objective functions, Zhao et al. (2012) derived a monotonic relationship between reservoir storage and optimal release decision under both deterministic and stochastic conditions, and proposes an algorithm to improve the computational efficiency of both deterministic DP and SDP for reservoir operation. The algorithm is used to solve water supply model using objective of minimization of period accumulated squares of the ratios of water shortage to water demand. Although formulating the reservoir operations as continues or even concave models can significantly reduce solution time and applicable for large scale reservoir systems, the state and decision discretizing based methods are still widely used, for it has no confines on objective functions and feasible regions, and need no approximation of the models. As long as the system scales are within the limits of solvable dimensionality, the state and decision discretizing based methods can be used to most of long term reservoir operations. Most existing solution methods are coarse global optimization or local optimization assuming only one optimum for each system state. Some researchers notice that there are near-optimal reservoir storage trajectories in DP models. Liu et al. (2011) examined the near-shortest paths method, the genetic algorithm, and the Markov chain Monte Carlo (MCMC) method, to explore multiple near-optimal solutions to reservoir operation problems. The literature show that even only the constraints of bounds on single reservoir storage and release are used, there are still possible multiple near-optimal solutions for the energy maximization model, and the methods more complex than standard DP or SDP are necessary to get near-optimal solutions. Rougé and Tilmant (2016) explored the consequences of, and proposes a solution to, the existence of multiple near-optimal solutions when using SDDP for mid or long-term river basin management. That there can be multiple local optima at a single state for non-concave maximization models may seriously confines the solution accuracy and causes unreasonable decisions. The multiple local optima problem was seldom emphasized in existing SDP models when deriving operating rules, possibly because that standard discretized DP is a global optimization method and is seldom restricted by objective function and feasible region characters of the reservoir operation models. While whether explicitly consider the multiple local optima problem or not in solving algorithms may affect the solution efficiency and quality, and this brings the demand of a new way to improve solution methods.

Cascaded hydropower reservoir operations must consider both water volume and water head for economic value maximization (Lund and Guzman, 1999). Power system demands can be formulated as constraints on power of single reservoirs or cascaded systems. This paper concerns the SDP model for energy maximization of cascaded reservoir operation in which the total power constraints intensified the non-concave characteristic and multiple local optima problem. Both quadratic and concave penalty functions are used to address the total power constraints. To improve solution accuracy, a two-stage solution algorithm combining traversing and search is proposed for quasi-global optimization. Starting from traversing over the feasible region, one or multiple promising points near local optima are identified, then a local search algorithm is used to obtain local optimizations near the points. The proposed methods are applied to cascaded reservoir system on middle-lower Lancang River. Operating rule surfaces and simulation results are obtained and compared for both kinds of penalty functions and for different algorithms including traversing, and two-stage algorithms of single point search and multiple point search.

2. Deterministic energy maximization model for cascaded reservoirs

A real-world reservoir operation can be very complex. It has to meet various demands without violating the constraints of the system. With the system wide total power constraints, deterministic energy maximization model in a finite time horizon is formulated as follows.

$$\text{Objective Function:}\quad \max E = \sum_{t=1}^{T}\sum_{m=1}^{M} p_m^t \Delta^t \tag{1}$$

Constraints:

$$\text{Water balance:}\quad s_m^{t+1} = s_m^t + 3600\Delta^t\left(I_m^t + \sum_{i=1}^{u_m} r_{U_{m,i}}^t - r_m^t\right) \tag{2}$$

$$r_m^t = q_m^t + d_m^t \tag{3}$$

Relationship among reservoir storage, turbine flow and power generation:

$$p_m^t = \eta_m q_m^t\{[z_m(s_m^t) + z_m(s_m^{t+1})]/2 - zd_m\} \tag{4}$$

$$\text{Initial and ending storage:} s_m^1 = si_m \tag{5}$$

$$s_m^{T+1} = se_m \tag{6}$$

$$\text{Storage bounds:}\quad \underline{s}_m^t \leqslant s_m^t \leqslant \overline{s}_m^t \tag{7}$$

$$\text{Maximum power and turbine flow:}\quad p_m^t \leqslant \overline{P}_m^t \tag{8}$$

$$q_m^t \leqslant \overline{q}_m^t \tag{9}$$

$$\text{Minimum total power:}\quad \underline{P} \leqslant \sum_{m=1}^{M} p_m^t \tag{10}$$

$$\text{Minimum reservoir release:}\quad \underline{r}_m^t \leqslant r_m^t \tag{11}$$

where E is the objective of energy generation in operating horizon; m, t are sequence number of reservoirs and periods respectively, $1 \leqslant m \leqslant M$ and $1 \leqslant t \leqslant T$; M is the number of reservoirs; T is the number of periods in operating horizon; p_m^t, $\overline{P}_m^t$ are the power generation and its upper bound of reservoir m in period t; Δ^t is the hour number in period t; η_m is power generation coefficient of reservoir m; q_m^t, $\overline{q}_m^t$ are the rate of turbine flow and its upper bound of reservoir m in period t; s_m^t, $\underline{s}_m^t$, $\overline{s}_m^t$ are the storage and its lower and upper bounds of reservoir m in period t; $z_m(s_m^t)$ is the reservoir level of reservoir m in period t, a function of s_m^t;

zd_m is the average downstream level of reservoir m; I_m^t is the rate of local inflow of reservoir m in period t; u_m is the number of immediate upper stream reservoirs of reservoir m, and $U_{m,i}$ is the sequence number of the ith immediate upper stream reservoir; r_m^t, $\underline{r}_m^t$ are the rates of release flow and its lower constraint of reservoir m in period t; d_m^t is the rate of spill flow of reservoir m in period t, in long term operation, constraint of its upper bound can be omitted; si_m, se_m are the initial and ending storage setting for reservoir m; $\underline{P}$ is the minimum total power, and is often set for power system demands.

In most cases, $z_m(s_m^t)$ can be fitted using polynomial functions. For some reservoirs with almost no storage capacity and nearly fixed water head (Madani et al., 2014), the degree of polynomial is 0 and E is a linear function of q_m^t according to Eqs. (1)–(4). When the degree of polynomial is 1, $z_m(s_m^t)$ is linear and E is a quadratic function of q_m^t. For many reservoirs, the degrees of polynomial are integer number no less than 2, and E is non-concave function of q_m^t. Moreover, the feasible region considering total power constraint of Eq. (10) may be a non-convex set. Taking a two reservoir problem at one no spill stage as an example, temporarily supposing linear reservoir level-storage curves as $z_m(s_m^t) = a_m s_m^t + b_m$ here, the upstream reservoir is numbered 1 and the downstream reservoir is numbered 2, so the constraint of Eq. (10) can be formulated as Eq. (12).

$$\underline{P} \leqslant \eta_1 q_1^t \left\{ \frac{(a_1 s_1^t + b_1) + a_1[s_1^t + 3600\Delta^t(I_1^t - q_1^t)] + b_1}{2} - zd_1 \right\} + \eta_2 q_2^t \left\{ \frac{(a_2 s_2^t + b_2) + a_2[s_2^t + 3600\Delta^t(I_2^t + q_1^t - q_2^t)] + b_2}{2} - zd_2 \right\} \tag{12}$$

Replacing q_1^t as x and q_2^t as y in Eq. (12), the boundary of feasible region is $\alpha x^2 + \beta y^2 - \beta xy + \gamma x + \lambda y = 0$, $x \geqslant 0$, $y \geqslant 0$, here $\alpha = -\eta_1 a_1 3600\Delta^t/2$, $\beta = -\eta_2 a_2 3600\Delta^t/2$, $\gamma = \eta_1\{[(a_1 s_1^t + b_1) + a_1(s_1^t + 3600\Delta^t I_1^t) + b_1]/2 - zd_1\}$ and $\lambda = \eta_2\{[(a_2 s_2^t + b_2) + a_2(s_2^t + 3600\Delta^t I_2^t) + b_2]/2 - zd_2\}$.

Relationship of Eq. (13) can be obtained through deriving the second derivative of Eq. (12).

$$-\frac{(2\beta y - \beta x + \lambda)}{2\alpha}\frac{d^2y}{dx^2} = \left(\sqrt{\frac{\beta}{\alpha}}\frac{dy}{dx} - \frac{1}{2}\sqrt{\frac{\beta}{\alpha}}\right)^2 + \left(1 - \frac{\beta}{4\alpha}\right) \tag{13}$$

The condition of the feasible region is non-convex is $\frac{d^2y}{dx^2} < 0$, and it can be satisfied in many circumstances. For example, where the product of generation coefficient and reservoir level-storage incremental rate of the downstream reservoir is less than 4 times of that of the upstream reservoir, then $\frac{\beta}{4\alpha} < 1$ and the right side of Eq. (13) is always positive. For $\alpha < 0$, the non-convex feasible region condition is $2\beta q_2^t - \beta q_1^t + \lambda < 0$. The condition may be satisfied when the optimal decision is in the region where q_1^t is greater than 2 times of q_2^t given small λ value. That often happens at dry season when the storage rate of the upstream reservoir is much more than the downstream reservoir. Local solutions exist in both regions of a and b in Fig. 1 given a non-convex feasible region, even for a concave objective function. The nonlinear $z_m(s_m^t)$ can cause much more complex feasible region, and either the $z_m(s_m^t)$ is formulated as piecewise linear or high degree polynomial curve, the non-convex feasible regions cannot be avoided.

So even the deterministic model may have multiple optima due to the non-convex feasible region caused by total power constraints, and can be intensified by non-concave objective function.

3. SDP model with soft constraints for cascaded reservoirs

3.1. Formulation

In order to derive operating rules for cascaded reservoirs considering uncertain inflow, infinite-horizon SDP models are often used to obtained rule tables to interpolate releases from reservoir states. The recursive function of SDP model for expected benefit maximization is

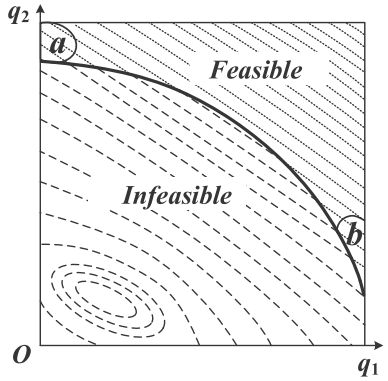


Fig. 1. Diagram of non-convex feasible region.

Eq. (14), and the constraints in the deterministic model except initial and ending storage remain for each inflow scenario. Common state variables include reservoir storage, previous-inflow and inflow forecasts (Stedinger et al., 1984; Kim and Palmer, 1997). In this paper, the reservoir storage is set to be state variables for easier comparison between different solution algorithms, and the conclusions can be extended to models with more state variables such as pre-inflow or inflow forecasting. With Eq. (14), the state transition function is same to Eq. (2) under each inflow scenarios.

$$f^{*t}(\mathbf{S}_k^t) = \max[F_{\mathbf{S}_k^t}^t(R_k^t)] = \max_{R_k^t} \sum_{n=1}^{N} \Pr(\mathbf{Q}_n^t)\{B^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) + f^{*t+1}[\mathbf{S}^{t+1}(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t)]\} \tag{14}$$

where $f^{*t}()$ and $f^{*t+1}()$ are the objective of period t and $t+1$, or future value function at the beginning of period t and $t+1$, respectively; $\mathbf{S}_k^t$ is the state variable: kth vector of discretized storages of reservoirs; R_k^t is the decision variable: vector of reservoir releases, at kth state; $F_{\mathbf{S}_k^t}^t(R_k^t)$ is the objective function of R_k^t at period t given state $\mathbf{S}_k^t$; $B^t()$ is the benefit function at period t; N is the number of combinations of discretized reservoir local inflows; $\mathbf{Q}_n^t$ is the vector of nth discretized local inflow of reservoirs, and $\Pr(\mathbf{Q}_n^t)$ is the probability of it; $\mathbf{S}^{t+1}()$ is the ending storage vector at period t, a function of the initial state, decision and inflow.

Due to uncertain inflows, the constraints on total power cannot be satisfied at all states under all inflows, so the constraint of minimum total power in the deterministic model (Eq. (10) becomes a soft constraint, and $\underline{P}$ is often called the firm power in stochastic operation models, denoted as P_f. The firm power and minimum release constraints are addressed using penalty functions in this paper. For the benefit lost from firm power failure is not accurately evaluated, the penalty function forms depend on willingness to accept different forms of shortage, continuous smaller shortage or a shorter period of greater shortage. This willingness can be decided by operators or managers knowing relations between shortage and system benefit or social aversion (Rossman, 1977) to shortage. The penalty augmented benefit function is formulated as Eq. (15).

$$B^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) = E^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) - g1^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) - g2^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) \tag{15}$$

In Eq. (15), $E^t()$ is the energy generation at period t, given storage and inflow vectors of $\mathbf{S}_k^t$ and $\mathbf{Q}_n^t$, following release decision of R_k^t. $g1^t()$ is the penalty function for firm power shortage, $g2^t()$ is the penalty function for minimum release shortage for non-power requirements (Shawwash et al., 2000) or social well-beings (Chang and Wang, 2013). The penalty terms are formulated as Eqs. (16) and (17).

$$g1^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) = \sigma \max\left(P_f \Delta^t - \sum\nolimits_{m=1}^{M} p_m^t \Delta^t, 0\right)^{\mu} \tag{16}$$

$$g2^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) = \sum_{m=1}^{M} \theta \max[3600\Delta^t(\underline{r}_m^t - r_m^t), 0]^2 \tag{17}$$

In Eqs. (16) and (17), $\sigma > 0$, $\theta > 0$, $\mu > 0$ are coefficients. When $\mu = 2$ Eq. (16) is a quadratic function, and when $\mu < 1$ Eq. (16) is a concave function.

If the system is expected to be operated for smaller extreme shortage of firm power, meaning that the unavoidable power shortages are allocated to several periods to make the largest single period shortage as small as possible, the quadratic function is used.

If the system is expected to be operated for fewer shortage periods, the constraint of firm power can be replaced by a constraint of failing chance. The demand can be formulated as the explicit chance constraint of Eq. (18),

$$\Pr\left(\sum_{m=1}^{M} p_m^t \geqslant P_f\right) \geqslant c \tag{18}$$

where c is the chance constraint on firm power.

Under this circumstance, an extreme minimum power constraint are needed to control the shortage extent, so there are two levels of total power constraints: firm power for normal periods and the extreme minimum power for extreme dry periods. The chance constrained model can be solved using reliability constrained DP (Rossman, 1977), and also can be approximately addressed using a concave penalty function for firm power failure with exponent less than 1 in Eq. (16). The extreme minimum power constraint can be addressed using a similar quadratic penalty function for firm power. The SDP recursive function is the same as Eq. (14) using the benefit function of Eq. (19),

$$B^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) = E^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) - g1^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) - g2^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) - g3^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) \tag{19}$$

where $g3^t()$ is the penalty function for the extreme minimum power constraint, and is formulated as Eq. (20), and σ in $g1()$ is changed to a value between 0 and 1. In Eq. (20), $\varphi > 0$ is a coefficient, P_e is the constrained value of the extreme minimum power, $P_e < P_f$.

$$g3^t(R_k^t, \mathbf{Q}_n^t, \mathbf{S}_k^t) = \varphi \max\left(P_e \Delta^t - \sum_{m=1}^{M} p_m^t \Delta^t, 0\right)^2 \tag{20}$$

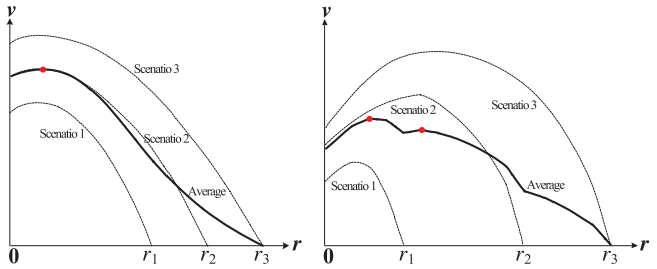
The concave penalty function tends to obtain results with fewer shortage periods comparing to the quadratic penalty function, while the extreme shortage can be larger. The quadratic penalty on extreme minimum power is used to confine the extreme shortage.

After addressing the constraints using penalty terms in objective function, the constraints remain at each state are only the upper bounds of turbine flows, and the feasible region of release is $[0, \bar{q}_m^t]$ for each reservoirs.

The non-concave characteristic of the model is intensified using penalty functions, for the energy generation is linear function of period power generation, while the penalty terms are power functions of period generation.

3.2. Multiple local optima problem of SDP for cascaded reservoir operations

Eq. (14) defines a recursive computation of periodic Markov processes in an infinite horizon, of which one cycle corresponds to a year, and each period is defined as one month. Su and Deininger (1972) show that the model converges to a steady release policy that maximizes the system's average annual benefit over an infinite horizon. The SDP solution needs to obtain optimal decision at each state from the last period T to the first period 1 backward in a year. This process is repeated iteratively until decisions converge at all periods and all states. At the beginning of each iteration, the future value function for period T is replaced by that of period 1 of the last iteration. The remaining problem is how to solve the sub-optimization models for optimal decisions at each state to maximize objective function of $F_{\mathbf{S}_k^t}^t(R_k^t)$. A traditional solution method for the sub-optimization is to simply traverse discretized decision combinations of reservoirs in the feasible region. The traversing method is a global optimization method because discretized decisions distributed in whole feasible region are tested, but the solution accuracy and computation effort depends on how dense the decisions are discretized.



(a) One optimum (b) Two optima

Fig. 2. Diagram of piecewise function summation.

For the sub-optimization models, besides the reasons of non-concave objective function and non-convex unpunished region similar to those for the deterministic model, another aspect that may enhance non-concavity and even intensify multiple optima is the summation of piecewise functions due to different inflow scenario. As the example shown by Fig. 2, for one reservoir, r_1, r_2 and r_3 are the release values that can emptying the reservoir for inflow scenarios 1, 2 and 3, respectively. The objective functions for scenario 1 and 2 are concave where r is less than r_1 and r_2, and are non-concave in region of $[0, r_3]$, due to value unchanged regions of $[r_1, r_3]$ and $[r_2, r_3]$. The average value function is also non-concave (Fig. 2(a)), and may have multiple optima (Fig. 2(b)).

Even the objective function $F_{(s_1^t)}^t[(r_1^t)]$ is not concave, the SDP model for energy maximization of a single reservoir has only one local optimum in most cases. The $F_{(s_1^t)}^t[(r_1^t)]$ curves may have segments where the function value does not change with decision when the ending storage reaches its bounds. This will not cause serious problems in solving, but confines the solving algorithm of steepest ascent from any point.

Although the following presented analysis and methods are suitable for larger scale problems, a two reservoir problem is used to describe the details for simplification. At some storage combinations, even the decision for a reservoir has a single local optimum given the other reservoir's decision, the decision combination may still have more than one local optima. An example surface of $F_{\mathbf{S}_t^k}^t(R_t^k)$ is shown as Fig. 3 for unconstrained model and models with firm power constraint addressed using quadratic and concave penalty functions, for low storages of two cascaded reservoirs, at the second last period T-1, in the first round of yearly backward recursion.

For the unconstrained model, the $F_{\mathbf{S}_t^k}^t(R_t^k)$ surfaces can have more than one near-concave areas, but the optimum can hardly be missed and the solution is often easy.

For the model with a total firm power constraint addressed using the quadratic penalty function, peak value surface of Fig. 3(b) demonstrates that there are two local optima at the two peak areas, and the two peak areas are separated by a gap caused by the crossing of two sub surfaces. The small gap can affect the solution because the peaks may be missed in traversing and the local search cannot move from one side of the gap to the other. If the peak area on the right side is missed in traversing, while the left side peak area is caught, the optimal solution can be considered to be on the left side. Even further search is executed from the optimal point identified, the real global optimum in right side can still be missed. For the concave penalty function (Fig. 3(c)) case, there are three local optima and the global optimum is easier to be missed than the quadratic penalty function. A straightforward performance of the missing global optima is that there are some unreasonable protrusions and sunken points in the resulting rule surfaces. With the backward recursion of SDP, the shape and space distribution of local optimal points are changed and more complex.

If the solving accuracy is low, many poor solutions may be obtained

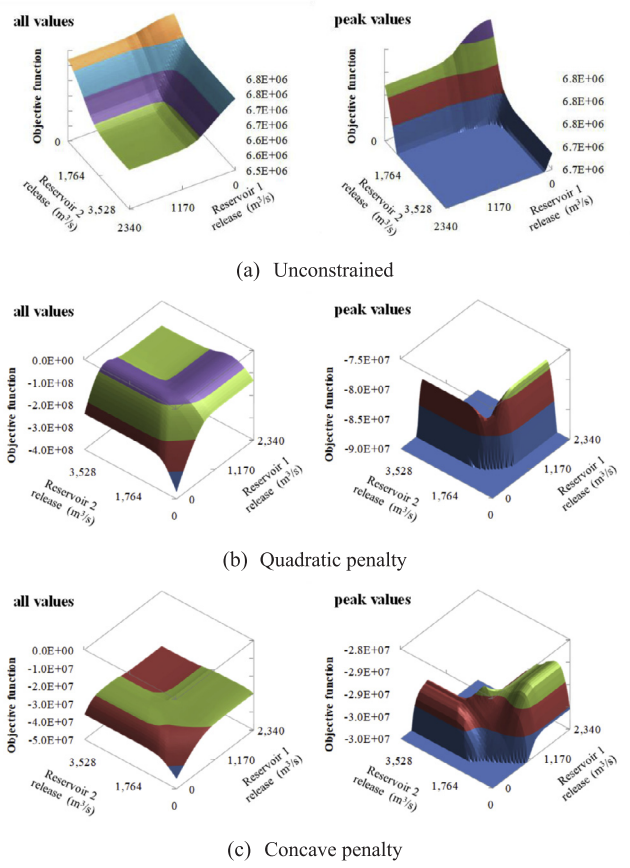


Fig. 3. Objective function surface examples.

for the multiple optima model. A poor solution will affect not only the current period decision, but also the future values of backward periods. Accumulated error on future values may cause ill-derived operating policies and even fail in solution. So it is necessary to solve the cascaded reservoir operation models using more effective global optimization algorithms.

4. Two-stage solving algorithm with local search strategy

Global optimization for multi-peak objective functions is an important and hard problem in engineering fields. There is not a universal method to obtain global optima for large scale multi-peak problems effectively. Although the DP algorithm has no restriction on the objective function, and can solve both concave and non-concave maximization problems, formulating the reservoir operation as a concave or near concave model is advantageous computationally to find global optima. When the problem can hardly be formulated as near concave models, finding an local optimum is relatively straightforward using local optimization methods, while finding the global optimization is far more difficult: analytical methods are frequently unavailable and numerical solutions are often computationally challenging.

Analyzing the solution of traversing, it can be noticed that when an optimal solution is obtained, better solution can also be expected nearby, and can be obtained through local search. Apparently search the region near the optimal traversed solution using smaller steps will increase the accuracy with less additional computation effort than dense discretization over whole feasible region. So the solution algorithm used in this paper at a storage combination includes two steps, traversing and search. For two cascaded reservoirs, the discretized storage point with sequence number of (x, y) is taken as an example. In the traversing stage, $L \times L$ evenly discretized points of release decisions in the feasible region are tested and objective function values are

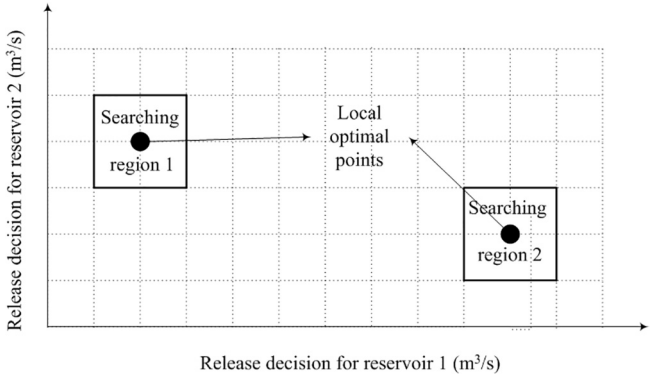


Fig. 4. Diagram of multi-point search.

obtained, let $i = 1{:}L$, $j = 1{:}L$ are the sequence numbers of discretized decision points for reservoir 1 and 2.

In the search stage, there can be two strategies. One is to search only the neighborhood of the optimal solution after traversing, called single point search. The other is to search all promising regions found based on the results of traversing, called multi-point search, as Fig. 4 shows. For the multi-point search, the discretized points near which local optima may exist are firstly found as initial search points. The initial search points are generated through three ways: (1) the region near optimal point in traversing stage is obviously promising, and the point can be an initial search point; (2) other discretized decision points with objective function values no less than all neighboring points, and at least larger than one neighboring point can also be initial search points. For point with sequence number (i, j), the neighboring points include points with sequence numbers of $(i + m, j + n)$, $m = -1, 0, 1$ and $n = -1, 0, 1$, and m and n are not 0 at the same time; (3) even the points found through (1) and (2) are saved, it cannot be ensured that all potential region having local optima are found, for the traversing points may not be dense enough to catch all the shape changing of the objective value surface, so some important points can also be set as initial search points. Considering that decision for neighboring discretized storage points may be close to each other in most cases, and a local optimum for a storage combination may exist near the solution of neighboring storage combinations, the optimal decisions of storage combinations with sequence numbers of $(x{-}1, y)$, $(x, y{-}1)$, and $(x{-}1, y{-}1)$ if exist, can be initial search points.

For there are platforms and crest lines on the objective value surface, to avoid too much redundant initial search points, the set of the initial search points is refined. For all initial search points that are neighboring one by one and have a same objective value, only one of them is retained.

For each initial search point, a local optimum is found by search. The search algorithm is important for it is called for thousands of times in one stage, and very little error may lead to an incorrect final solution. Set the ith initial search point as (a_1^i, a_2^i), the area of the rectangle with vertexes of $[a_1^i \pm a_1^{\max}/(L{-}1), a_2^i \pm a_2^{\max}/(L{-}1)]$ is defined as the control area of (a_1^i, a_2^i). Setting initial search steps as δ_m, $m = 1{:}2$, the search algorithm is to find optimal point in the points of $(a_1^i + j \times \delta_1, a_2^i + k \times \delta_2)$, j = -1,0,1, and $k = -1,0,1$. The search is repeated until no better solution is found. Then set $\delta_m = \delta_m/2$ and repeat the search for several times until preset minimum search steps for reservoirs are reached. The search algorithm from one promising point is shown in Fig. 5. The search algorithm can find a neighboring local optimum with very high precision, and can obtain solution outside the control area of the initial search point in some cases (Fig. 5(b)).

After the local searches, optima near all initial search points have been found, and the best of them is set to be the solution.

In solution, the interpolation on $f^{*t+1}()$ can be obtained by fitting the surface near point of (s_1^{t+1}, s_2^{t+1}) to $as_1^{t+1} + bs_2^{t+1} + cs_1^{t+1}s_2^{t+1} + d$, using four points around it. Although the multidimensional linear

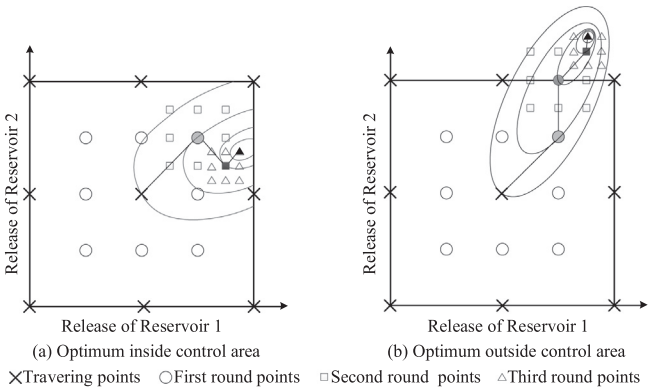


Fig. 5. Diagram of local search method.

interpolation or spline interpolation methods are more often used, the 4 variable-4 coefficient function used here can ensure that all points used in fitting obtain the real value on the fitted surface, and can fit both concave and convex surfaces with very small error.

Fig. 6 shows the computing flow of the algorithm of multi-point search, and the computing flows of the other two algorithms of only traversing and single-point search are similar to it except that the solution at each state are simpler.

The two-stage solution methods consume more computing time than traversing, but the complexity and computing effort are not much increased. So the extreme solvable reservoir number is close to common SDP.

5. Case study

5.1. Experimental setting

The Lancang River is an international river, known outside China as the Mekong River. The Lancang-Mekong River is about 4900 km long and flows through China, Myanmar, Laos, Thailand, Cambodia and Vietnam before discharging into the South China Sea. The hydropower system studied in this paper consists 5 cascaded reservoirs on middle-lower Lancang River, in Yunnan Province. The characteristics of the studied hydropower reservoirs are shown in Table 1. The hydropower system has installed capacity of 14820 MW and more than $221 \times 10^8 m^3$ effective storage capacity, making it the important seasonal regulation power resources of its service regions. In the service regions, there are many other hydropower reservoirs have no long term regulation ability, and the energy generation of them in flood season is much larger than in dry season. The seasonal regulation of the studied system can avoid further increasing of quantity gap of hydroelectricity between flood season and dry season, achieved through the firm power constraints in the operation model to move the flood season energy to dry season. Storage capacities of Xiaowan and Nuozhadu reservoirs are much greater than the others of the system, so the storage of the other reservoirs can be neglected and their reservoir levels are set near the upper bounds of different months. The percentages of extreme power generating head falling to maximum generating head of Xiaowan and Nuozhadu reservoirs are near 30% and 23% respectively, so the long term head variations play an important role in operation. Monthly inflows data used are from year 1953 to year 2008.

The solution algorithms are programmed in the Java language. The SDP models with 51 discretized storage points are tested. Local inflows of Xiaowan and Nuozhadu reservoirs are correlated to each other, and 5 local inflow intervals for each reservoir are used to get discretized joint probability distributions at each period. Local inflows to Manwan reservoir and Dachaoshan reservoir are supposed to be frequency synchronous to Xiaowan reservoir inflows, and local inflows to Jinghong reservoir are supposed to be frequency synchronous to Nuozhadu

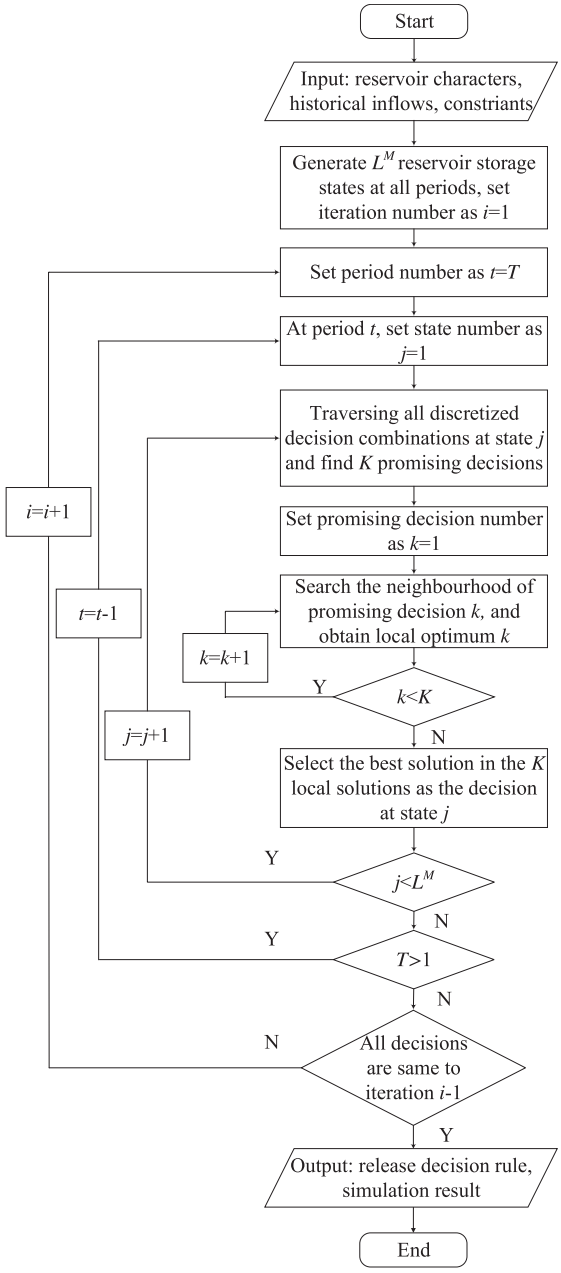
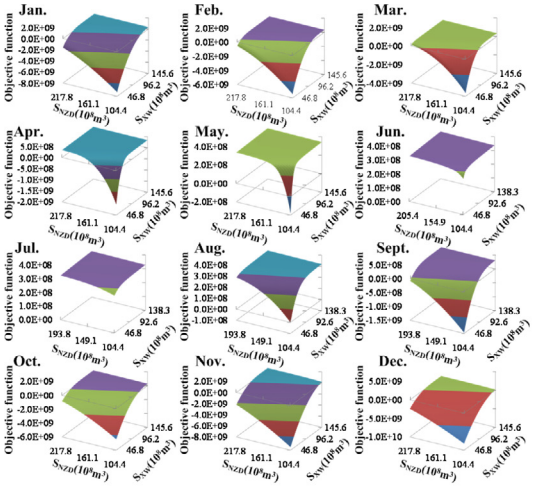


Fig. 6. Computing flow of multi-point search.

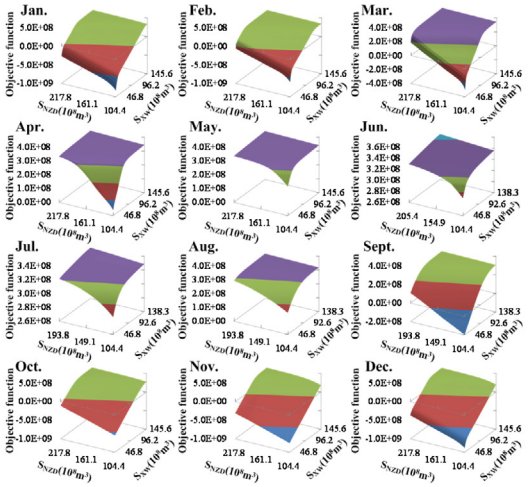
Table 1
Characters of Hydropower Plants on middle-lower Lancang River.

Plants	Installed capacity (MW)	Effective storage capacity (10^8 m^3)	Power generating water head variation (m)
Xiaowan	4200	98.77	174–248
Manwan	1670	2.57	81–93
Dachaoshan	1350	3.70	67–84
Nuozhadu	5850	113.35	161–208
Jinghong	1750	3.09	53–64

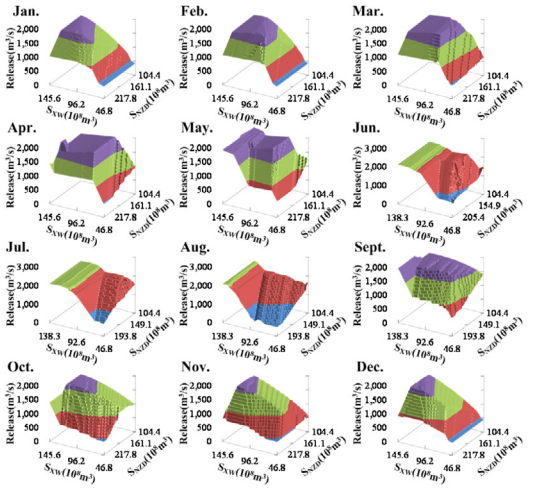
reservoir inflows. The minimum releases of Xiaowan, Nuozhadu and Jinghong reservoirs are set as $470 m^3/s$, $500 m^3/s$ and $504 m^3/s$, respectively. Parameters of the model using quadratic penalty function for firm power failure are $P_f = 6800$MW, $\mu = 2$, $\sigma = 10^2/(24*31)^2$, $\theta = 10^4/(24*31*3600)^2$, 10^2 and 10^4 penalty levels are used to reflect priority for different constraints, and other constants are used to



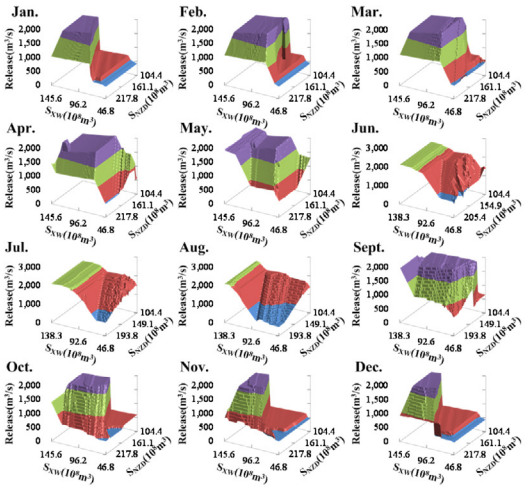
(a). Objective value surfaces using quadratic penalty function



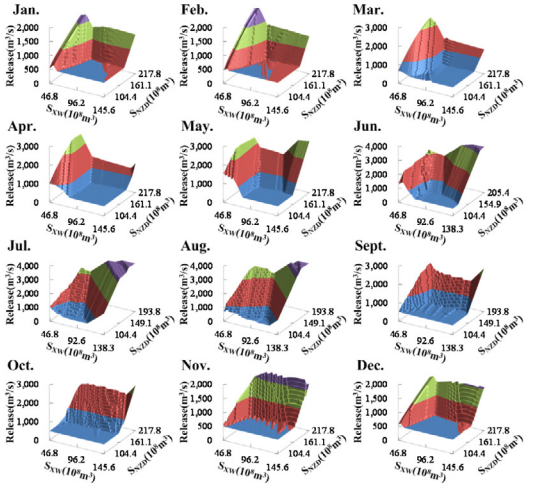
(b). Objective value surfaces using concave penalty function



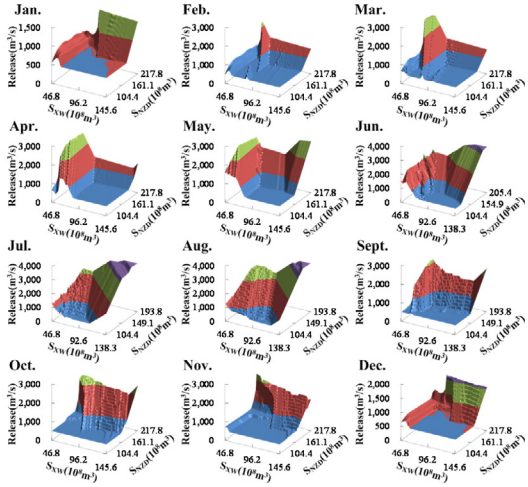
(c). Xiaowan rule using quadratic penalty function



(d). Xiaowan rule using concave penalty function



(e). Nuozhadu rule using quadratic penalty function



(f). Nuozhadu rule using concave penalty function

S_{XW}-Storage Xiaowan, S_{NZD}-Storage Nuozhadu

Fig. 7. Objective value and decision surfaces.

normalize the hour numbers in different periods. Parameters of the model using the concave penalty function for firm power failure are $P_f = 6800\text{MW}$, $P_e = 3000\text{MW}$, $\mu = 0.9$, $\sigma = 10^5/(24*31)^\mu$, $\varphi = 10^2/(24*31)^2$, $\theta = 10^4/(24*31*3600)^2$.

The models are solved by methods of traversing, single point search and multi-point search using different numbers of traversing points respectively. In search, the two dimension search method is used with initial search step as half of the traversing intervals and reduced by half for 8 times.

5.2. Numerical results

Fig. 7 compares the results of objective function and rule surfaces obtained using quadratic and concave penalty functions for firm power failure, multi-point search and 21 discretized release decision points in traversing for each reservoir. The major difference between the two operating rule sets is the shape in the dry season, when the penalty for firm power failure affects the results significantly. The rule surfaces using the quadratic penalty function are gradually changed from where the current period firm power demand can be fully satisfied to where failure may happen. The gradual changes cause nearly average shortage distribution among several periods if shortage cannot be avoided. With a concave penalty function, there are rapid changes in both reservoir rule surfaces. The rapid changes often mean the switches from satisfying firm power to satisfying extreme minimum power or minimum release. In both penalty functions, the upstream Xiaowan reservoir takes major task of regulation for total power and downstream flow, and the downstream Nuozhadu reservoir tends to stay high level unless storage of Xiaowan reservoir is lower than certain levels in different months. In most period of flood season, operating rule surfaces are similar using different penalty functions, for the constraints on firm power can be fully satisfied and the reservoirs try raise water head and avoid spill. Fig. 8 shows the simulation performances of the rules using quadratic and concave penalties.

Table 2 shows the results using different types of penalty functions and solution algorithms. Although in different cases, the iteration times on time horizon of SDP are different when stable decision obtained, the results after 5 times iteration are used here to compare the objective function values. In iterations after 5 times, the rule surfaces changed very little. The computing time is estimated for the computations with different settings are performed in different computers, and the time consuming need to be prolonged or reduced according to the computer configurations in order to comparing.

The operating rule using the quadratic penalty function on firm power failure obtains larger minimum total power and greater failure chance than using the concave penalty. The minimum power is confined by P_e using concave penalty function on firm power failure, and the simulated minimum total power is near P_e. In both penalty functions, the multi-point search algorithm obtained the best results, while the results obtained using only traversing are the worst, evaluated using the average objective values of January, or the average future values at the beginning of January. The combined traversing and search method can confirm the solving accuracy achieved by only traversing using very dense discretized points. Even 101×101 decision points are traversed consuming 30 times more time, the average objective value still cannot achieve that of the simplest two-stage solving method: single point search using 11×11initial decision points. Using the multi-point search algorithm with 21×21 initial decision points, the computing time is about 50% and 12% of that using only traversing algorithm for 51×51 and 101×101 points respectively. So to obtain operating rules with similar performances, the two-stage algorithms can save at least 50% computing time comparing to only traversing.

Using only traversing, the average objective values increase by 3.5% and 31.3% comparing using 11 discretized release decision points to using 21 discretized release decision points, for quadratic and concave penalties, respectively. That indicates the density of traversed points affects the concave penalty function model more seriously than the quadratic penalty function model. For the quadratic penalty, the single point and multi-point searches increase the average objective value by 1.3% comparing to only traversing using 21 discretized decisions. In concave penalty cases, the increasing percentages are 13.0% and 13.3%. For two-stage algorithms, the increasing percentage using concave penalty is larger, illustrating that the concave penalty function models are influenced by multiple local optima more seriously than that of quadratic penalty, and need higher solving accuracy. The average objective values of two-stage algorithms change very little with traversing point density, illustrating the two-stage algorithms are more robust to the traversing point density.

When a simple operating rule is used, in which the cascade tries to generate the firm power of 6800 MW, by operation of Xiaowan and Nuozhadu reservoirs for same ratios of ending storages to the maximum storage constraints at all periods, 3812.9TWh energy generation and 3016 MW minimum total power are obtained. So the SDP models increase energy generation by 0.5%-0.9% comparing to the simple operating rule.

Most part of the rule surfaces should be smooth or near smooth for

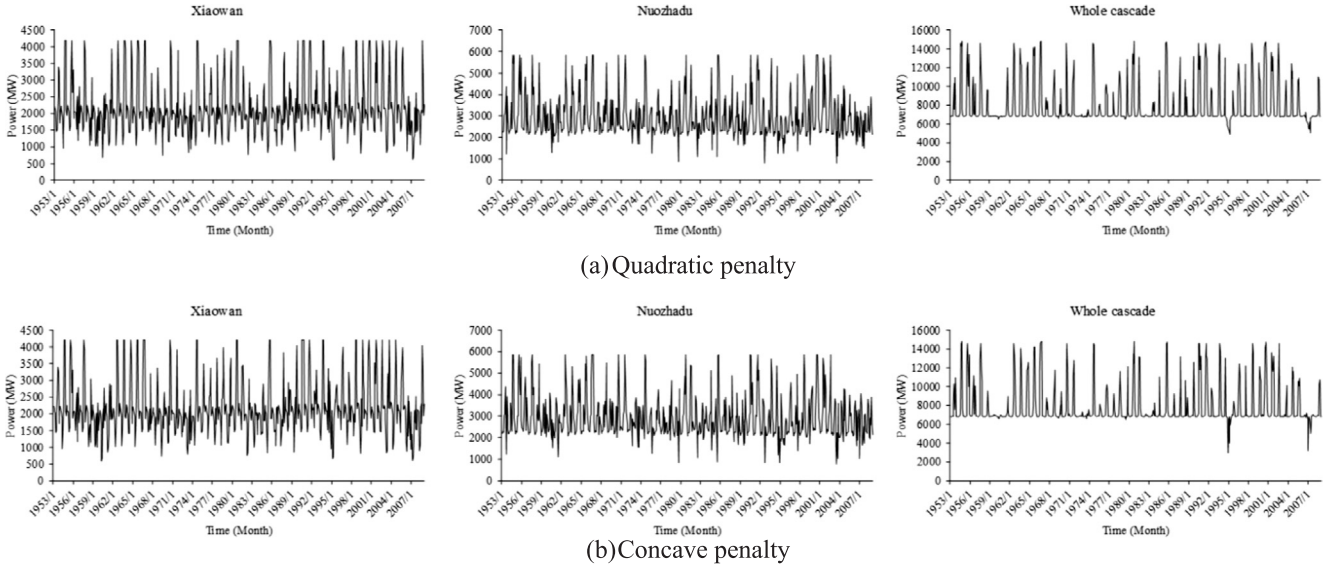


Fig. 8. Simulation performances.

Table 2
Computing results.

Penalty type for firm power failure	Quadratic							Concave						
Solving Method	Multi point		Single point		Only traversing			Multi point			Single point		Only traversing	
Traversing point number	11	21	11	21	11	21	51	101	11	21	11	21	11	21	51	101
Total energy (TWh)	3848.5	3848.4	3848.4	3848.2	3835.7	3844.4	3847.0	3848.3	3846.7	3846.8	3846.0	3846.1	3831.3	3842.1	3844.2	3844.8
Minimum power (MW)	4871	4873	4872	4873	4885	4852	4884	4807	2943	2944	2940	2942	3121	3096	2956	2952
Reliability, 10 MW tolerance (%)	85	85	85	85	87	88	89	86	96	96	96	96	93	96	96	96
Average value function ($1\,0^{8}$) of January after 5 iterations on time horizon	−8.0795	−8.0794	−8.0805	−8.0804	−8.4720	−8.1854	−8.1070	−8.0900	0.8973	0.8977	0.8942	0.8951	0.6029	0.7919	0.8696	0.8826
Estimated computing time on a Pc with 3.0 Ghz CPU (hour)	45	62	15	30	8	23	124	500	45	62	15	30	8	23	124	500

small changes of states often leading to small decision variation. All the rapid changes in the surfaces should represent a certain operating intention. Without an explicit operating intention, the rapid changing, especially the sunken or protruding points are unreasonable and very possible caused by numerical problems. Although the final average objective function and the performance of multi-point and single point search are close to each other, the obtained rule surfaces can be quite different in some states. Fig. 9 show the differences of rule surfaces of February for both penalty types using different solution methods. The difference of the release decisions for both reservoirs can be easily found. There are more sunken points in the decision surfaces of Xiaowan reservoir and accordingly protrusions in decision surfaces of Nuozhadu reservoir, comparing the result using traversing and single point search to multi-point search. The sunken or protruding points cannot be explained by operating demands but solving accuracy, especially as better solutions are missed in traversing and single point search. Even the unreasonable decision points did not cause considerable differences in both the average objective value and the simulation performance using observed data, they may influence application of the rule and result in hardly explained decisions using more inflow scenarios in future. Multi-point search can avoid most such circumstances, and obtain rule surfaces with more accuracy and acceptability, but with larger computation effort than the other two algorithms.

In fact, the more the discretized number of states is, the more precise the result can be. While the computing time will increased exponentially with discretized numbers. The discretized numbers used are enough to this problem which can be reflected by numerical results in that the simulating performances are good, and can only be very slightly improved through increasing discretized numbers. The penalty coefficients are not optimized, and used to show the effect of the solution methods. In fact, the coefficients need to be optimized to find values that can best balance the energy generation and the constraint on total power and reservoir release. But that is very time consuming: solution time may be several hours to several days depending on the solution algorithm and traversing density. When larger coefficients are tested, the simulation results show no significant improvements in reliability, which indicates that the coefficients used are large enough for this case study.

Like other SDPs, the proposed method can be extended to three reservoirs, but the inflow correlations need to be simplified and the discretized state and decision numbers cannot be too large to avoid too much computing effort. There are many cascaded hydropower reservoirs that consist no more than 3 large reservoirs having long term regulation ability and more small reservoirs can be seen as run-of-river stations. For multiple cascaded hydropower reservoir systems, the near optimal operation decision can be made through two step: allocating cross basin constraints, such as the firm power for whole system, to each cascade, and making single cascade operation decisions. So even having dimensionality confines, the proposed model can still be widely used. Also, the methods can be integrated into aggregation based SDP algorithms or SSDP solution for models having more reservoirs, and more complex time and space inflow correlations.

6. Conclusions

The multi-optima problem of reservoir operations are often caused by non-concave objective functions and non-convex feasible region, and can be enhanced by summation of piecewise functions due to different inflow scenario in SDP. To illustrate this problem, both deterministic and stochastic energy maximization models are proposed. In the stochastic model, quadratic or concave penalty functions are used to address total power demands. To solve the SDP models for cascaded systems with multiple local optima to obtain the optimal decision at each discretized storage state combination, a two-stage global optimization algorithm combining traversing and search is proposed in which a local search algorithm is applied to single or multiple promising

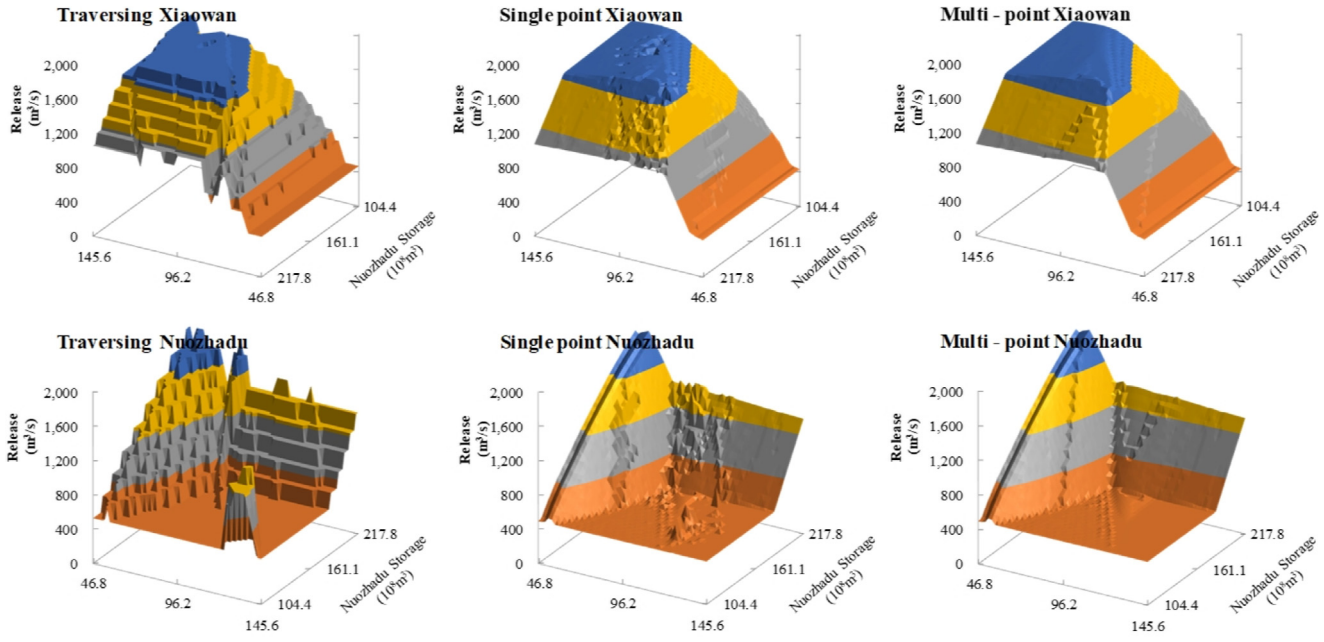


Fig. 9. Comparison of rules by different method for February.

regions found through coarse traversing. Case study for reservoirs on middle-lower Lancang River in China shows that: (1) the quadratic and concave penalty can reflect preferences for smaller shortage and fewer shortage periods, respectively; (2) local search after traversing is much more effective than dense discretizing, and can obtain improved solutions with both better objective values and simulated performance in much shorter computing time; (3) although the results of multi-point search has no significant improvement on average objective values and simulation results, it is superior to single point search for more reasonable and acceptable rule surfaces with fewer strange decision points; (4) the number of discretized point of decisions in one step can significantly affect the results of traversing, but have smaller impacts on the two-stage solution algorithms; (5) the proposed method obtain joint operating rules of cascaded hydropower reservoir system with two large reservoirs and additional smaller reservoirs in acceptable computing time, running on PCs, and can be combined with reservoir aggregation based methods and inflow sampling methods for larger scale and more complex problems.

Acknowledgments

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